Model Predictive Control for Minimizing HVAC Systems' Energy Consumption and Peak Load without Sacrificing Thermal Comfort

Daniel Newell, Kamran Vakil, Ruiji Sun, Shuo Zhao

***Abstract*—We studied the design of Model Predictive Control (MPC) for Heating, Ventilation, and Air-Conditioning (HVAC) systems. Our MPC controller's main objective is to reduce HVAC systems' energy consumption and peak heating or cooling power without sacrificing occupants' thermal comfort. Based on perfect thermodynamics model knowledge, we implemented our MPC on the same model as the one used for simulation. We analyzed how the controller reacts to disturbances in occupants' load and outside air temperature. The results show that MPC will perform significant preconditioning behavior if either the HVAC systems power or electricity peak load is limited. Additionally, a large building thermal mass can strengthen the preconditioning effect. https://youtu.be/FHbjTrsbnkw**

I. Introduction

One of the HVAC systems' primary objectives is to maintain the indoor environment within a comfortable temperature range; however, researchers at the Center for the Built Environment (CBE) found that 89% of buildings do not qualify the 80% occupants' thermal satisfaction requirement [1]. As the popularity of Smart Buildings increases due to the Internet of Things (IoT), thermal comfort has become a bigger priority in the built environment. Modern HVAC systems utilize controllers with short cycle inputs in order to maintain a reference temperature in buildings. This method allows for more straightforward implementation in buildings, at the expense of worse thermal comfort and more considerable energy waste. HVAC systems account for more than 50% building energy consumption in developed countries [3]. Furthermore, these controllers fail to utilize predictions in their computations, meaning external factors such as outside temperature, occupancy, and other disturbances which can be accounted for by IoT sensors can continuously influence the system.

II. System model identification

This project aims to first apply the MPC controller to a simplified building thermodynamics model instead of much more complex models shown in research papers. The MPC robustness can then be strengthened by introducing more complex models and disturbances. We use a Resistance-Capacitance (RC) circuit analogy [2] to model the function of a house heat transfer balance:

$$C\dot{T} = u + d + \frac{T^{oa} - T}{R} \tag{1}$$

T is the indoor air temperature. T^{oa} is the outdoor air temperature and u is the HVAC systems heating or cooling power, depending on whether u is positive (heating) or negative (cooling). d represents additional power input created by occupants' loads. R represents the lumped thermal resistance of the entire house, including walls, windows, and other barriers between the inside space and outdoor environment. C represents the lumped thermal capacitance of the entire house, including indoor air, furniture and the building envelopes. Then we discretize the linear continuous time system using an Euler discretization. T_k is the indoor air temperature at time k and Δt is the time step. In this paper, we will use the superscript as variables and subscript as time. Therefore:

$$T_{k+1} = T_k + (\frac{u_k}{C} + \frac{d_k}{C} + \frac{T_k^{oa} - T_k}{RC})\Delta t \tag{2}$$

$$T_{k+1} = AT_k + Bu_k + Bd_k + (1 - A)T_k^{oa} \tag{3}$$

where

$$A = 1 - \frac{\Delta t}{RC}, B = \frac{\Delta t}{C}$$

We used the example MATLAB thermal model of a house for simulation and the RC model parameter estimation. The original MATLAB model did not include building thermal mass, so we added an equivalent air thermal mass into the model as shown in Figure 1. Average outdoor air temperature was set as 12 °C and daily temperature variation followed a sine wave. Thermostat temperature set point was 20 °C with 1 °C band. The heat transfer delay then became significant due to the large thermal mass effect, shown in Figure 2. The simulation period was 24 hours, and the time step was 1 second. MATLAB outputs of outdoor air temperature, indoor air temperature and HVAC heating power were exported into Python and used to solve the parameter estimation problem.

$$\min_{A,B,e} \sum_{k=1}^{N} ||e_k||_2^2 \tag{4}$$

subject to

$$e_k = T_{k+1} - AT_k - Bu_k - Bd_k - (1 - A)T_k^{oa} \tag{5}$$

$$\begin{bmatrix} e_1 \\ \vdots \\ e_N \end{bmatrix} = a\begin{bmatrix} A \\ B \end{bmatrix} - b, \tag{6}$$

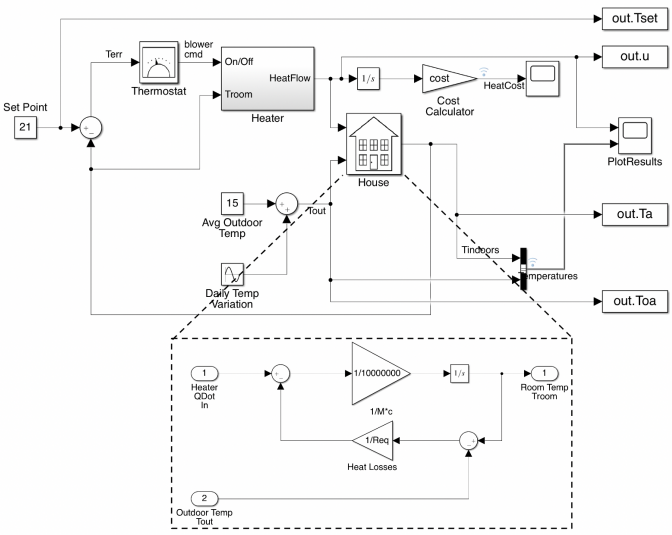


Fig. 1. MATLAB thermal model of a house

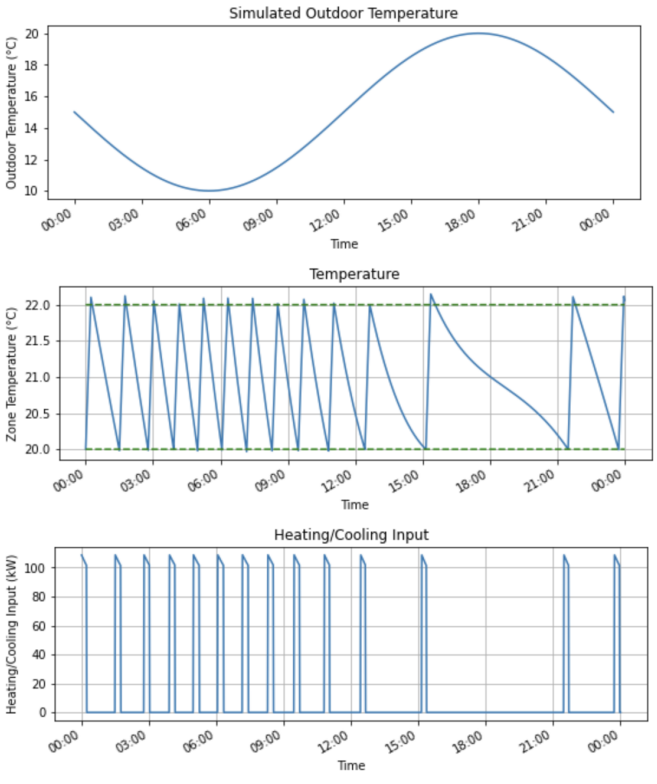


Fig. 2. MATLAB simulation result

The model estimation error e_k can be minimized by proper choice of parameter a and b. They were populated using output data from the MATLAB simulation. N is the number of total data samples used. e_k is 1 step model estimation error of the indoor air temperature. $a \in \mathbb{R}^{N\times 2}, b \in \mathbb{R}^N$. Then, quadprog was used to solve this problem. The results are $A = 9.99\text{e}{-01}$ and $B = 2.78\text{e}{-04}$ for only air thermal mass as $1.00\text{e}{+06}$; $A = 1.00\text{e}{+00}$ and $B = 2.78\text{e}{-05}$ for combined air and building thermal mass as $1.00\text{e}{+07}$; $A = 1.00\text{e}{+00}$ and $B = 2.78\text{e}{-06}$ for combined air and building thermal mass as $1.00\text{e}{+08}$. Estimated data created by this RC model, perfectly matches with the MATLAB simulation data. Therefore, the system model used for our MPC can be assumed same as the one used for simulation.

III. Cost Function Design

The MPC controller was designed to minimize overall energy consumption, peak power load and variations from the comfortable temperature range. At each time step, the predictive controller obtained the optimal input sequence satisfying constraints, while minimizing the following objective function:

$$\min_{U_k,\overline{\varepsilon},\underline{\varepsilon}} \sum_{k=0}^{N-1} |u_k|\Delta t + \kappa max(|u_k|,...,|u_{N-1}|) + \rho \sum_{k=1}^{N} (|\overline{\epsilon}_k| + |\underline{\epsilon}_k|) \tag{7}$$

subject to

$$T_{k+1} = AT_k + Bu_k + Bd_k + (1-A)T_k^{oa}, \tag{8}$$

$$\underline{T} - \underline{\varepsilon}_k \leq T_k \leq \overline{T} + \overline{\varepsilon}_k, \tag{9}$$

$$\underline{\varepsilon}_k, \overline{\varepsilon}_k \geq 0, \tag{10}$$

$$-50 \leq u_k \leq 50, \tag{11}$$

where $|u_k|$ is the magnitude of the heating or cooling input at each step. The optimal zone temperature range is denoted by $\underline{T}$ and $\overline{T}$ for the lower and upper bounds, respectively. $\underline{\varepsilon}$ and $\overline{\varepsilon}$ are the values of the violations outside of the optimal temperature range at each step. Any T value outside of $\underline{T}$ and $\overline{T}$ will result in a higher cost. The heating and cooling input is hard constrained in (11) to a magnitude of 50kW due to the physical limitations on our heating/cooling source. The cost function is weighted using κ and ρ.

For our simulation, $\underline{T}$ and $\overline{T}$ were set to 20 and 22 °C. The initial zone temperature was set to 22°C to better show the precooling effect when a heat disturbance in introduced. The controller horizon N was set to 200, which allows the controller to optimize over the upcoming 3 hours 20 minutes at each step. We assumed there is no model mismatch between the simulation and our MPC model.

Outdoor air temperature T^{oa} and occupancy disturbances d were calculated to show the efficacy of the controller. The simulation ran for 24 hours. For the simulation without a sinusoidal outdoor air temperature, the outdoor temperature was set to a constant 22°C. The disturbance profile was implemented as an occupancy load that takes place between 14:00

and 16:00. The sinusoidal outdoor air temperature represents a more realistic outdoor temperature scenario over the course of 24 hours and oscillates between 15°C and 35°C.

IV. Results

A. Small Load

Within the context of a smart building, scheduled personnel can be treated as disturbances. We model this 20kW disturbance such that it is within our input's energy limits. The result, as depicted in Figure 3, demonstrates the benefit of MPC's future predictions. By foreseeing the incoming disturbance, the controller lowers the model's temperature to 21.3°C. This precooling effect allows the system to remain within its preferred temperature bounds while minimizing energy usage. In addition, it allows the controller to react to a disturbance with a lower peak energy demand.

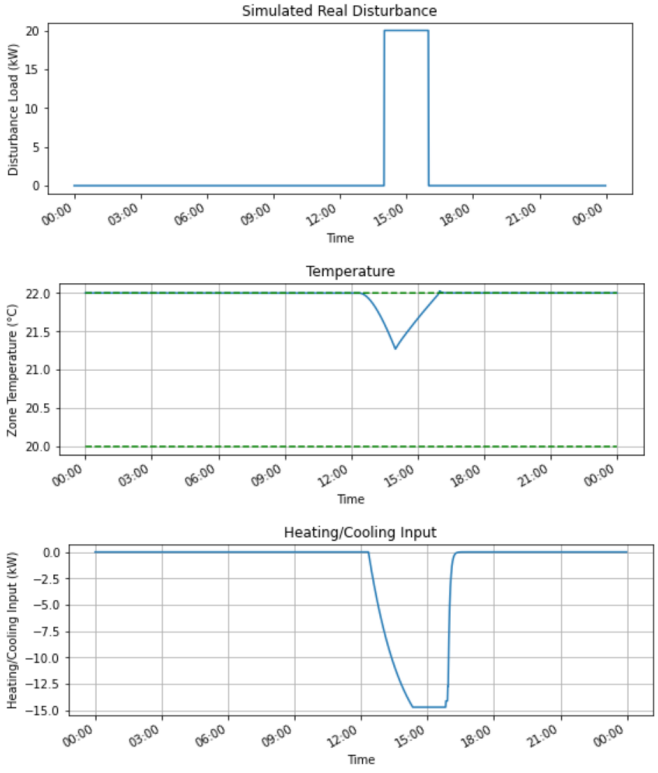


Fig. 3. MPC control with small load disturbance.

B. Large Load

This simulation, as depicted in Figure 4, includes a larger disturbance analogous to a large group of people entering a building for a conference.In this case, the 100kW disturbance is much larger than the input's maximum cooling capability. Due to the magnitude of this load and 50kW limits on the inputs, the building is precooled lower than 20°C in anticipation of the upcoming spike in temperature. For a contemporary controller, this would be impossible to properly cool. Indeed, even our MPC controller would be unable to continue supporting the large disturbance if it persisted indefinitely. However, our controller can keep the building temperature within its preferred bounds for longer periods of time than contemporary controllers. This is once again due to the controller cooling the system before the disturbance is applied, effectively prepping our building for the influx of people.

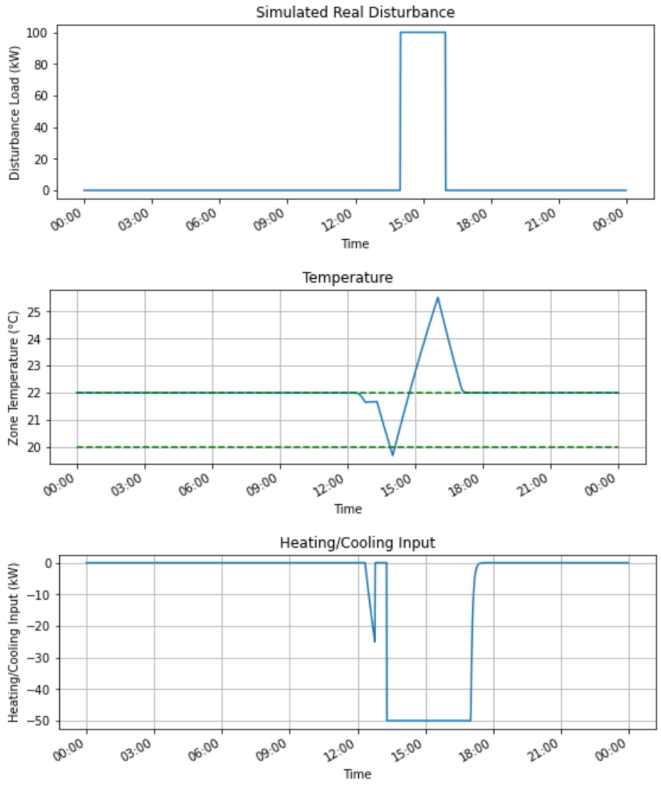


Fig. 4. MPC control with large load disturbance.

C. Large Load with unlimited input

This simulation, shown in Figure 5, removes the hard constraints of the input while leaving the minimization of energy input and peak energy input intact. This simulation also demonstrates the efficiency of the MPC controller, as it only requires a peak power of 83kW to deal with a 100kW load. Compared to the constrained input simulation, the cooling input is used for a shorter time span due to its ability to put out over 30kW more power. This extra power allows the controller to keep the zone temperature within the comfort range for almost the entire simulation.

In contrast, a tuned PID controller, shown in Figure 6, is incapable of keeping the plant within temperature bounds even with a peak power input of 100kW. Further tuning of the controller could yield different results for our cooling system, however none of these results would be capable of keeping our temperature within range for the duration of the large load disturbance when the energy input is limited. This is because the implemented disturbance will be larger than our input. Without the input limit, the PID controller uses a

large peak energy input to bring the temperature back to its preferred range. Another more aggressive PID controller might be able to lower the temperature violation by increasing the proportional gain, however this result would be analogous to a short cycle control method which is already implemented in contemporary HVAC systems. Furthermore, increasing this gain too much could cause system instability.

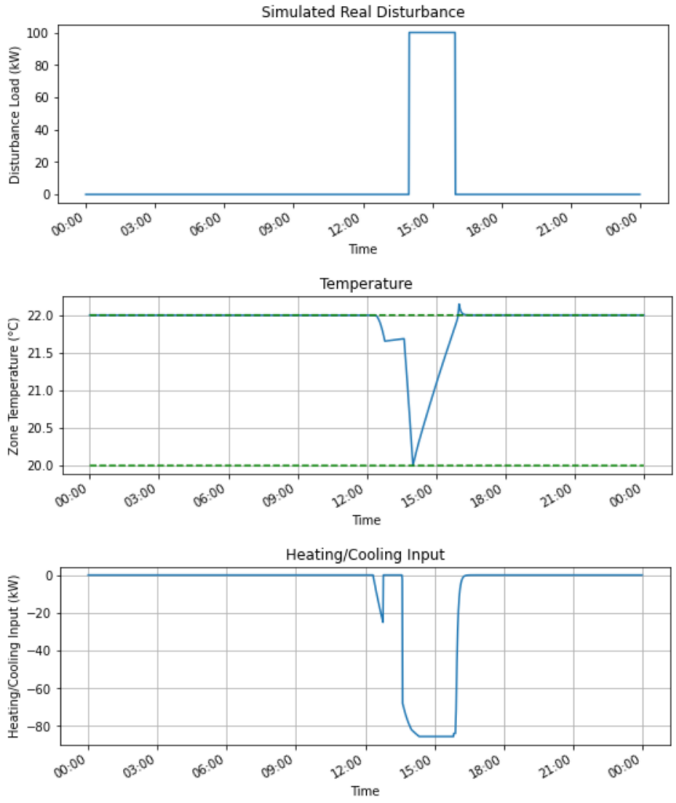


Fig. 5. MPC control with large load with unlimited input.

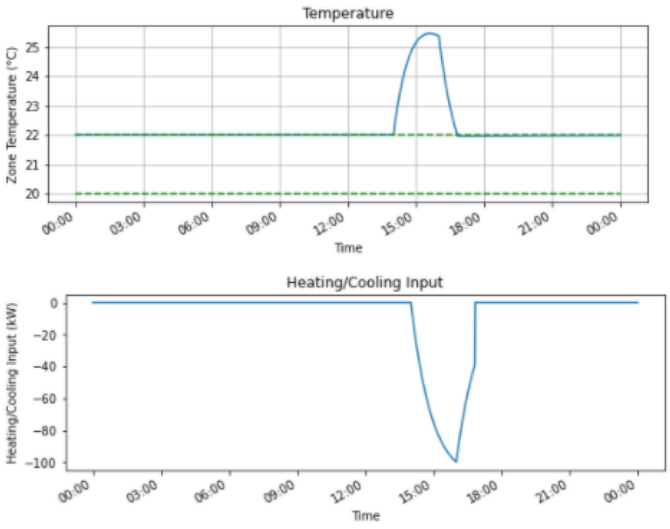


Fig. 6. PID Control with large load with unlimited input

D. Simulated Outdoor Temperature Disturbance

This simulation, depicted in Figure 7, shows the efficiency of the MPC controller when dealing with a sinusoidal outside temperature profile instead of a constant 22°C temperature profile like the previous cases. No short-term disturbance load is included in this simulation. The controller is able to keep the temperature within the comfort range for the entire simulation. These results show that the MPC controller is capable of controlling the building's temperature for standard outside temperature fluctuations over the course of 24 hours.

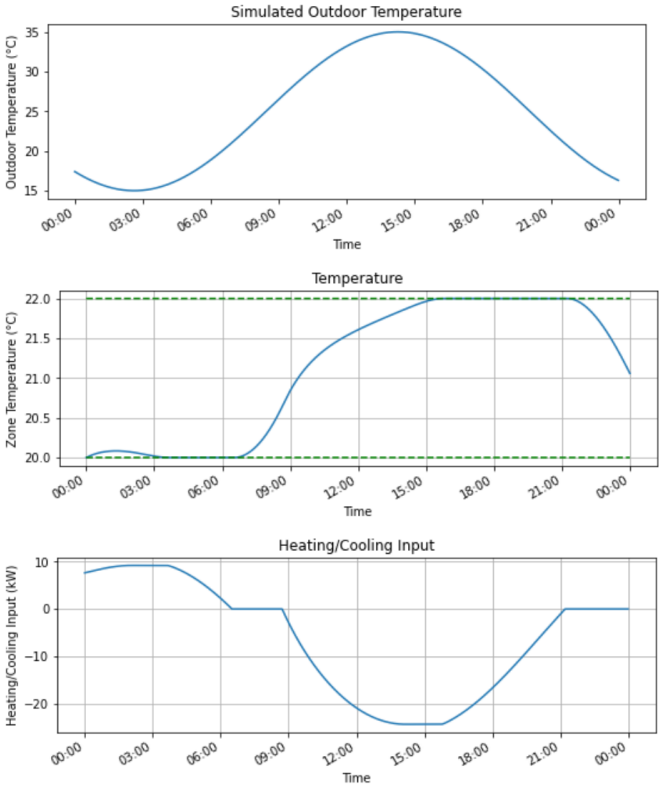


Fig. 7. MPC control with Simulated Outdoor Temperature Disturbance.

E. Simulated Outdoor Temperature with Small Load Disturbance

This simulation, shown in Figure 8, shows the most realistic scenario for this controller where outside temperature fluctuates and a small load disturbance of 20kW is included. As compared to the previous simulation in Figure 7, the input curve follows largely the same path initially, but then a small amount of precooling is instituted at around 13:00 to account for the upcoming small load. By having full knowledge of upcoming loads, these results show that this controller is capable of precooling to account for predicted upcoming loads even when outdoor air temperature is included as a disturbance. These results suggest that the controller could be best implemented with weather data and a calendar that includes future room usage in order to best predict the time and size of future loads and precool accordingly.

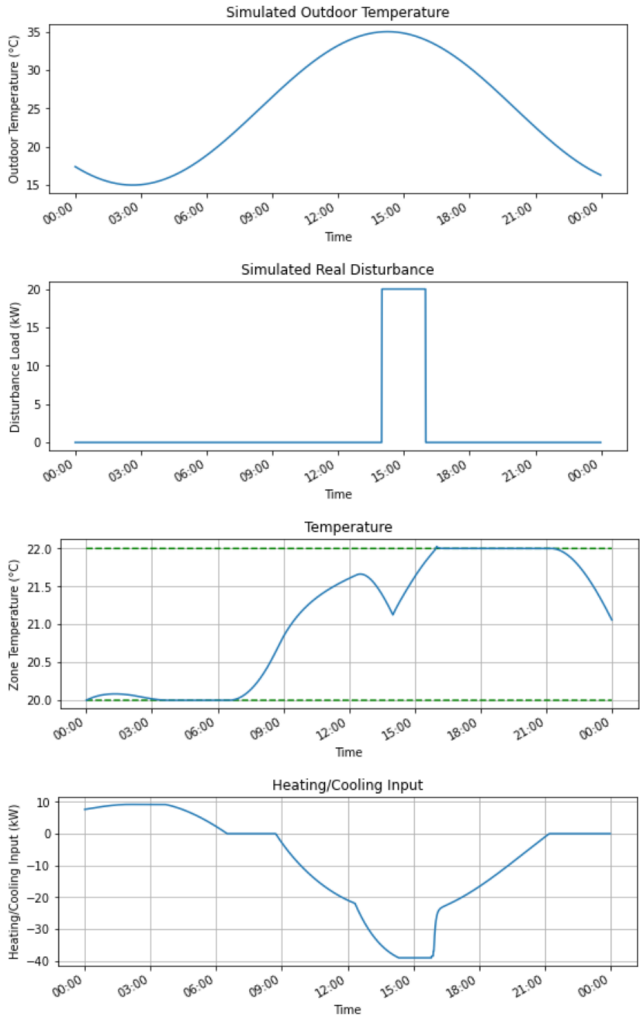


Fig. 8. MPC control with Simulated Outdoor Temperature and small load disturbance.

Conclusion

Traditional HVAC systems utilize controllers with short cycle inputs to maintain the desired temperature of the building, which generalize huge energy consumption and fail to adjust working period to fit changes from external factors. In this paper, a Model Predictive Controller (MPC) is designed to reduce the energy consumption of HVAC system and utilize predictions of external factor changes to have better control performance. A simple thermal dynamic model is constructed for MPC simulation, and building thermal mass is added into this model in order to better fit the real conditions. By implementing the MPC, the controller exhibits good performance in pre-heating and pre-cooling to predict the incoming disturbances and successfully allows the system to maintain the desired temperature with minimum energy consumption.

References

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